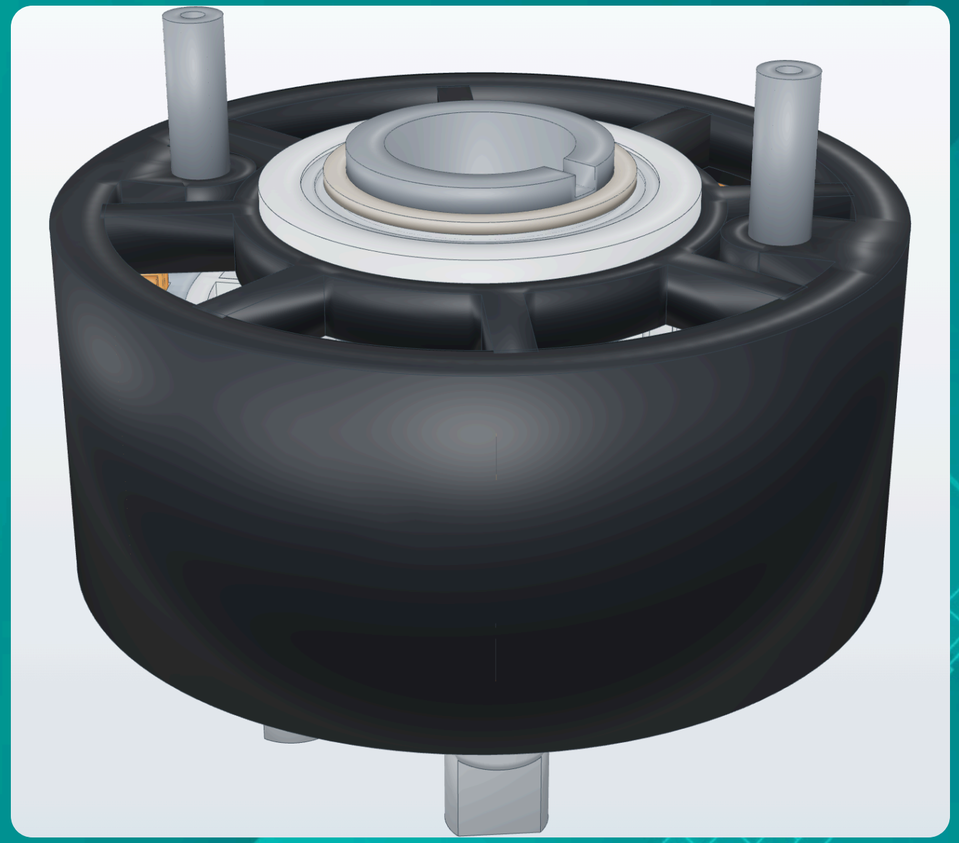
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AL-SULTAN ABDULLAH

EMDLAB · ELECTROMAGNETIC DESIGN LABORATORY

Drone motor optimisation

Teardown, 3D scanning, simulation and material optimisation of the DJI Agras T40 motor: a capability case study.



UMPSA AEROSPACE

Prepared for UMPSA Aerospace

October 2026

Kejuruteraan • Teknologi • Kreativiti

One pipeline, from a motor on the bench to an optimised rebuild

01

Teardown and measure

Disassemble, weigh and photograph; resistance and drill back-EMF tests.

02

3D scan

Shining 3D FreeScan UE Pro captures housings and mounts at 0.3 mm.

03

Parametric CAD

One parameter file drives STEP, STL, drawings and a 3D web viewer.

04

Electromagnetic FE

Nonlinear FE with rotor motion: back-EMF, torque, iron loss, demagnetisation.

05

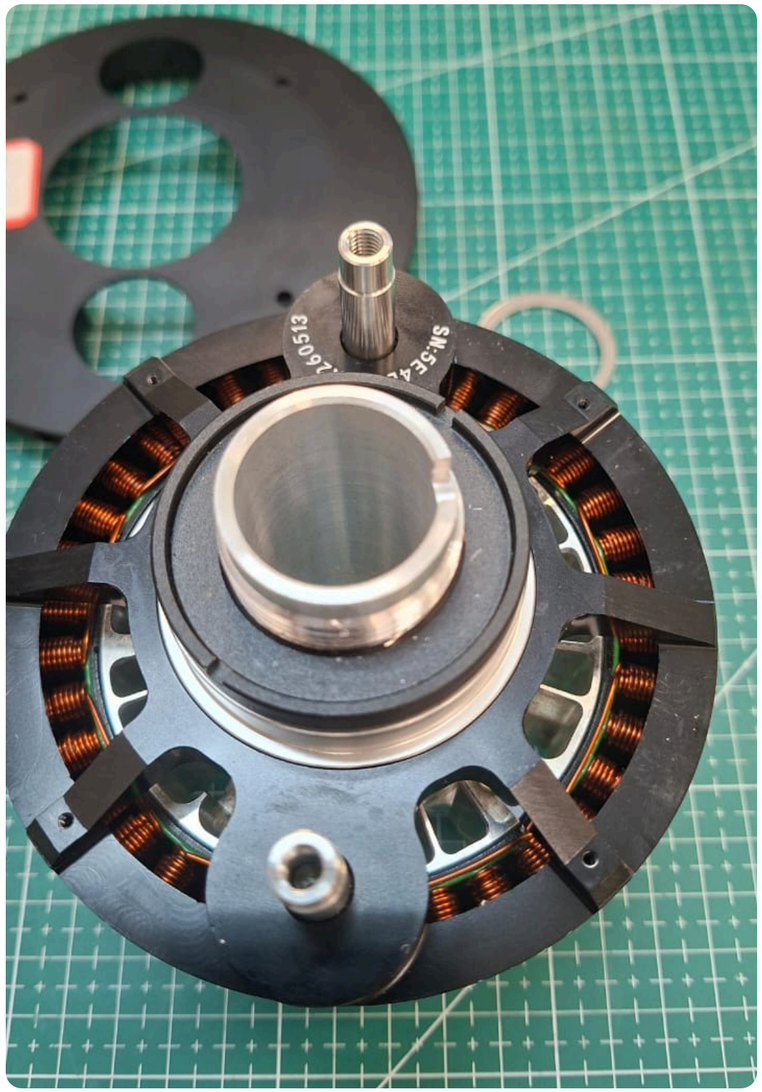
Thermal network

A 13-node transient model run over any race or mission profile.

06

Optimise and rebuild

Screen materials, pick builds, print and assemble, then bench-test.



CASE STUDY

The DJI Agras T40 propulsion motor

A 4 kW outer-rotor motor from a commercial agricultural drone: a demanding, real benchmark for the method.

Item	Value
Type	Outer-rotor PM motor, 36 slots / 42 poles
Stator	100 × 33 mm; motor OD 112 mm
KV	48 rpm/V
Resistance	22 mΩ line to line
Rating	100 A, 4 kW for 180 s on 14S
Mass	1250 g

Reseller specification, not a DJI datasheet; our bench values replace it as they come in.

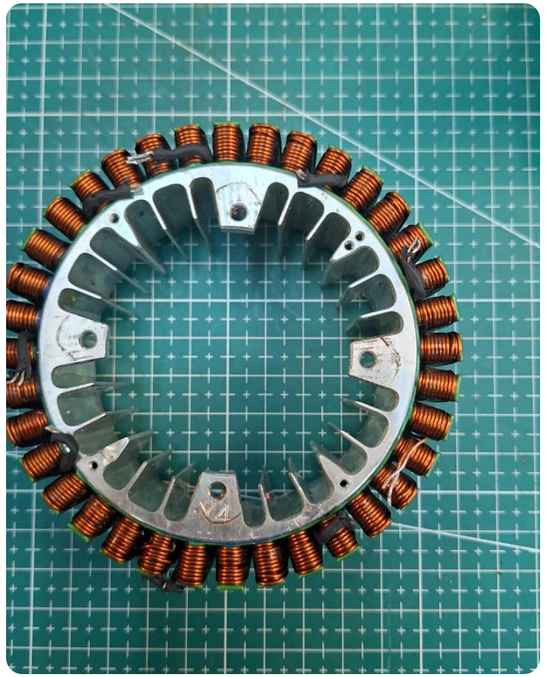
Taking the motor apart, and putting it back together



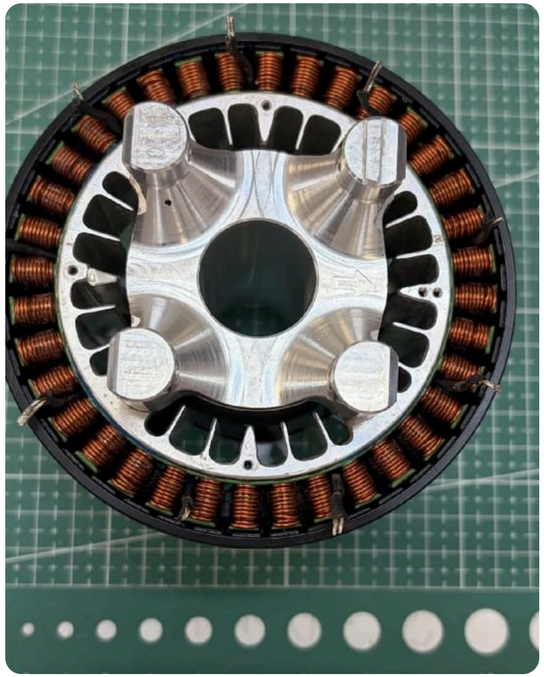
Video: fixed shaft and X-mount into the stator



Video: bearing and rotor onto the stator

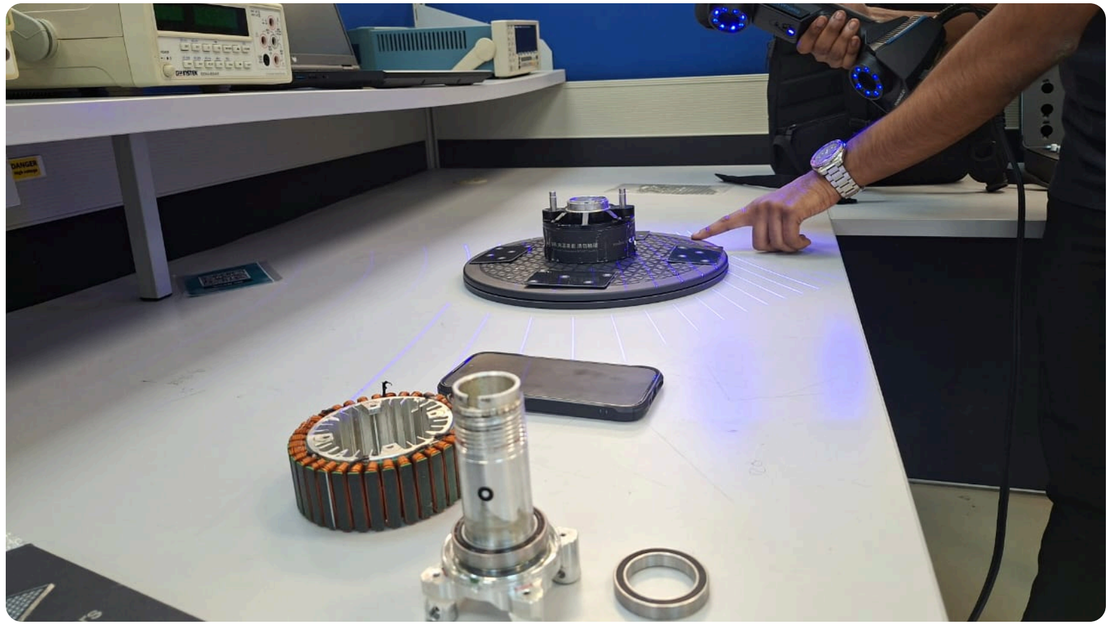


Finned aluminium stator carrier

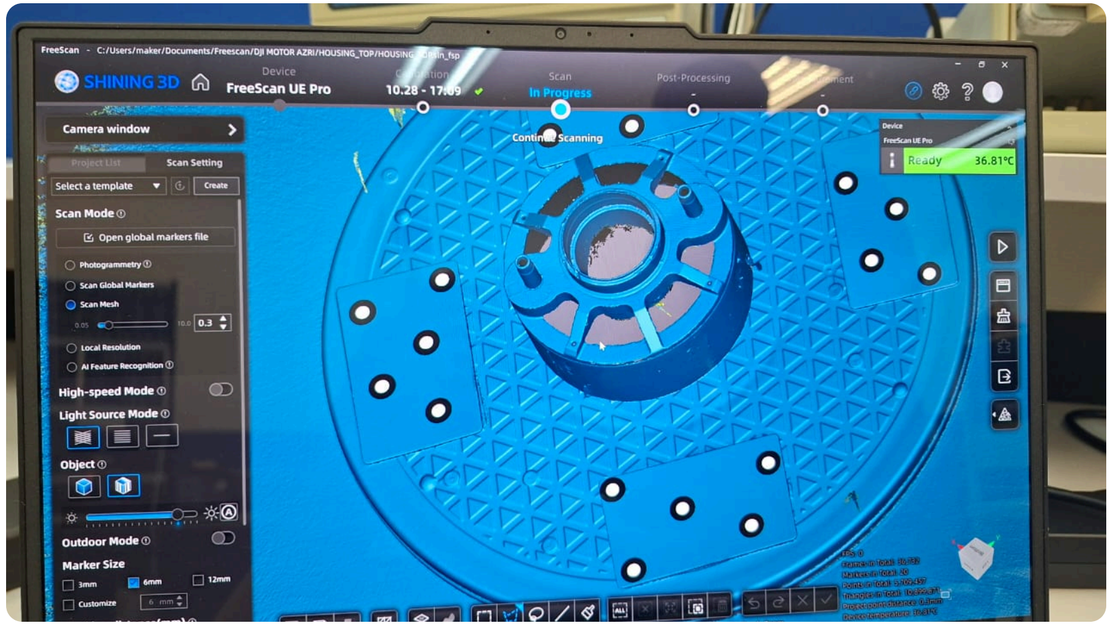


X-mount with four airframe posts

Scanning what calipers cannot measure



Handheld laser scanning of the rotor housing on a marker turntable



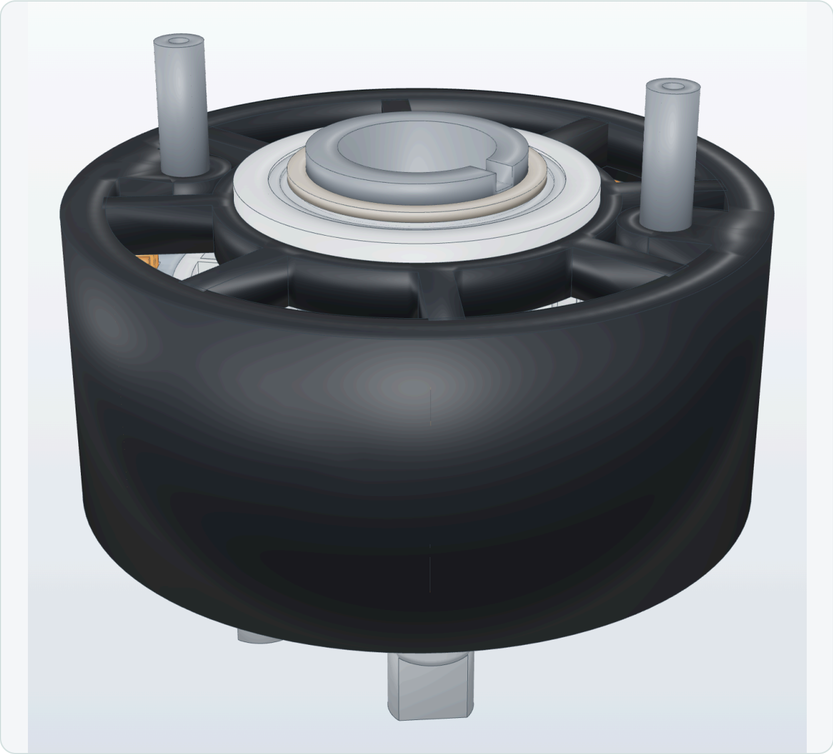
Live mesh of the spoked cap in Shining 3D FreeScan

FreeScan UE Pro, 0.3 mm point spacing

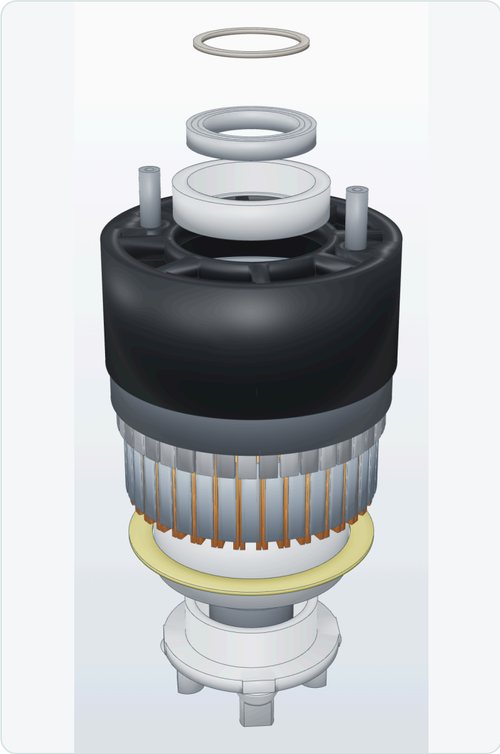
Scan → CAD fit: housing, cap, mount

Feeds the parameter file directly

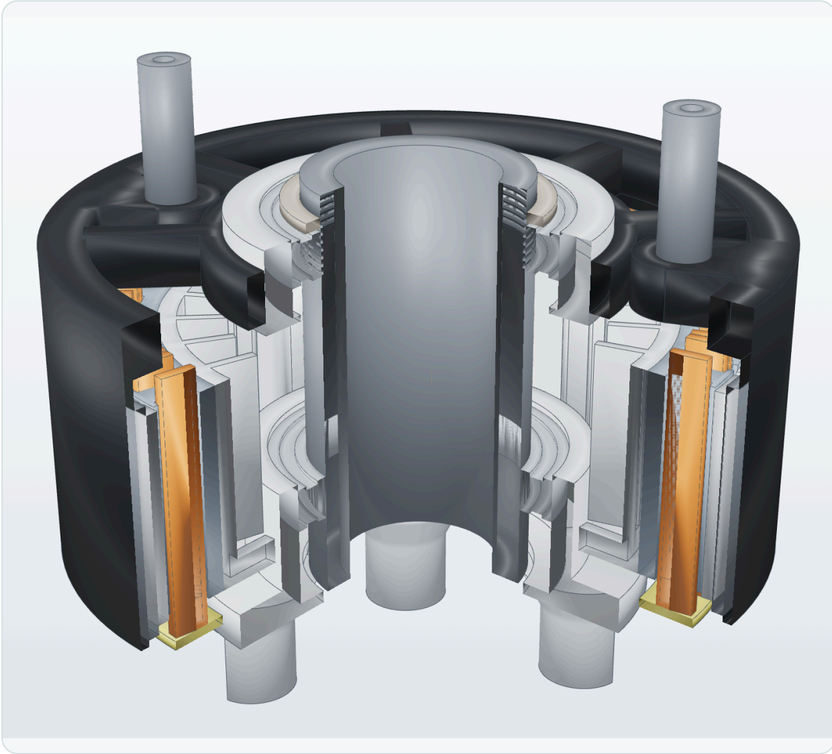
The motor rebuilt as an editable model



Assembly: black can, spoked cap, two prop pins



Exploded: 18 named parts



Section: coils, magnets, carrier, axle, bearings

THE OPTIMISATION QUESTION

A race motor that runs at its limit for one race, then gets replaced

Geometry fixed

Only materials change: laminations, magnets, enamel, structure.

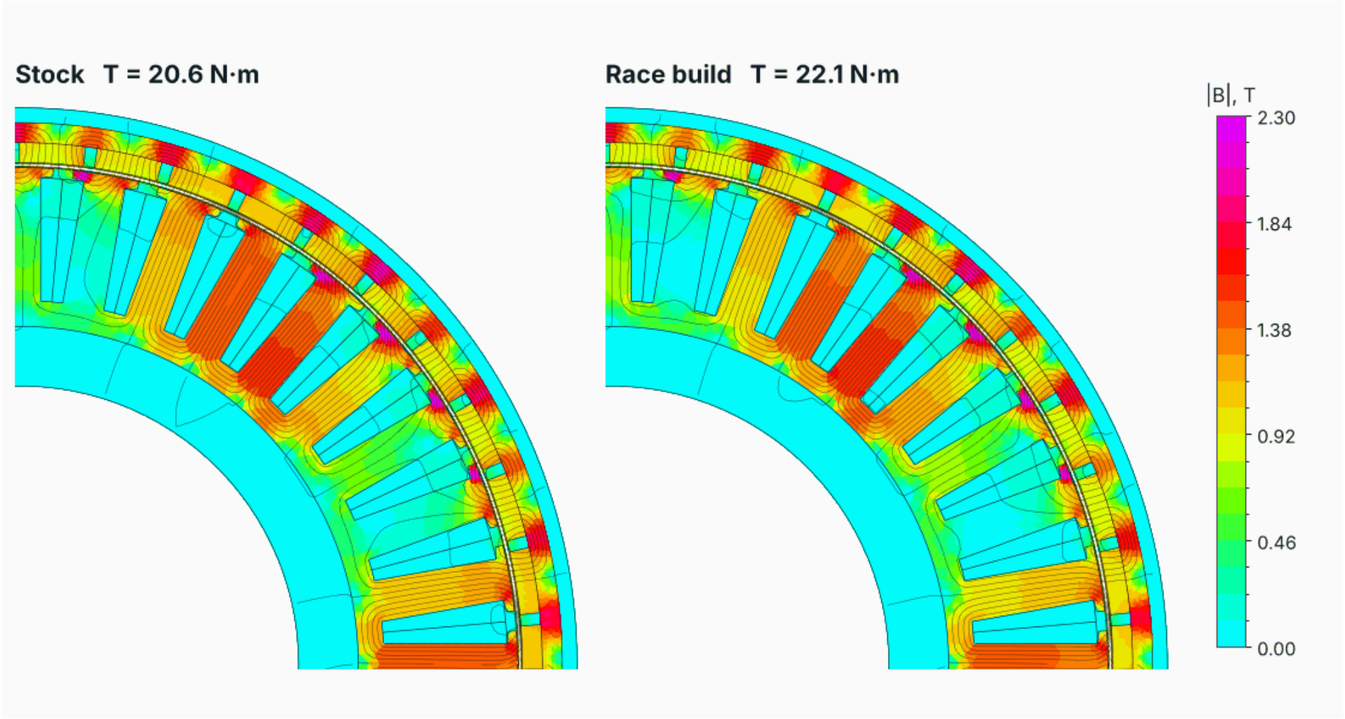
Race length open

Any profile from 30 s sprints to continuous, as the circuit decides.

Single-race limits

Enamel, magnet demagnetisation, polymer softening, stress.

The real field over one electrical period

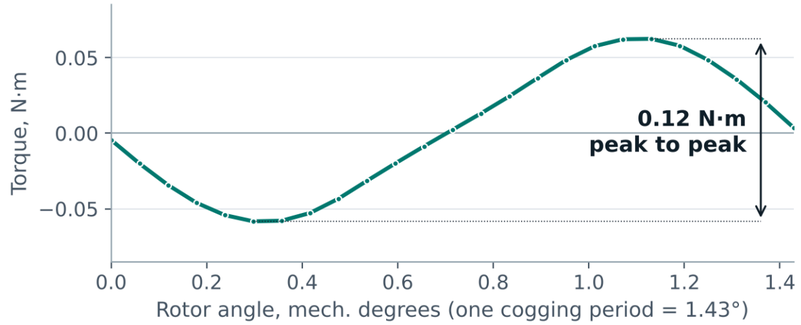


|B| in tesla on a quarter of the motor at the 180 s race current

48.0 rpm/V

FE gave 52; calibrated to the spec with one magnet-strength factor (×1.11)

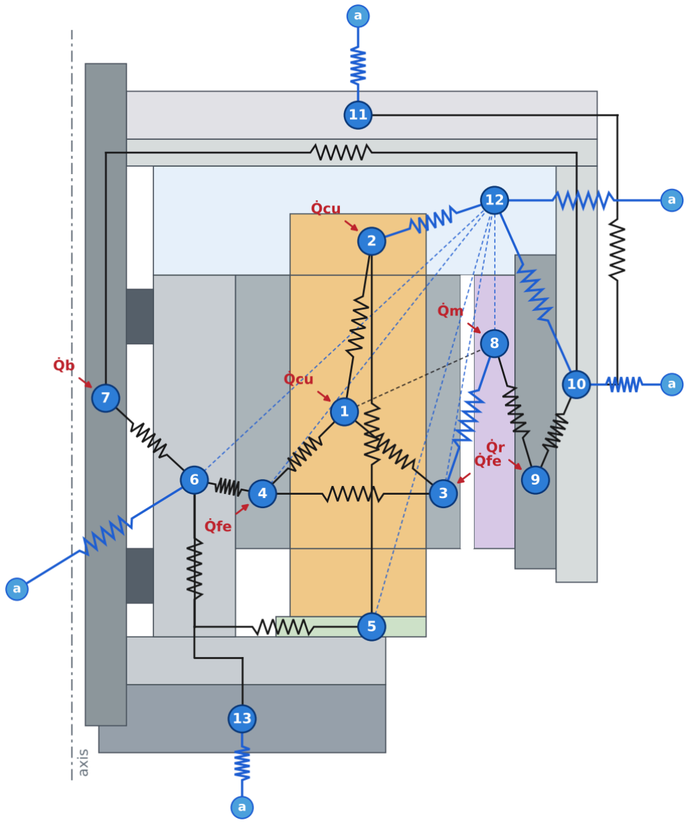
Cogging torque, 0.7 % of rated torque



2.2 T

tooth tips saturate

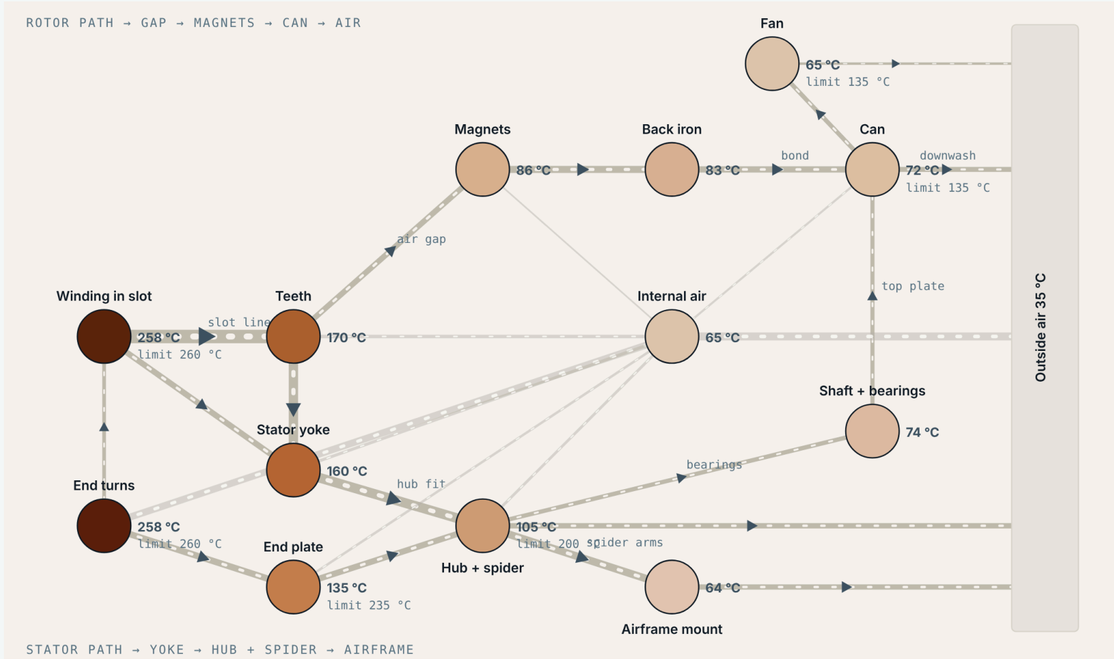
A 13-node thermal network mapped onto the motor



Nodes

1. Winding, slot
2. End turns (hot spot)
3. Teeth
4. Yoke
5. End plate
6. Finned carrier + X-mount
7. Fixed shaft + bearings
8. Magnets
9. Back iron
10. Rotor housing: can wall
11. Rotor housing: spoked cap
12. Internal air
13. Airframe mount
a. Ambient air

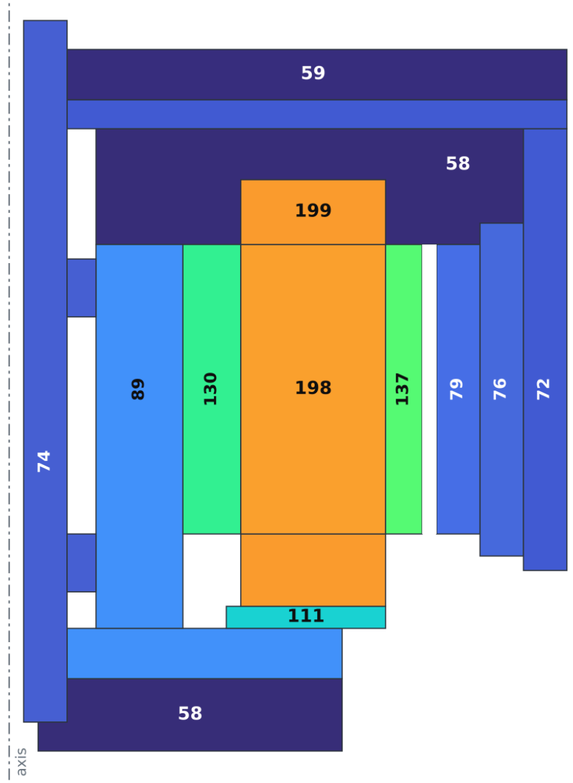
Conduction
Convection (speed dependent)
Loss source
Minor link (< 0.4 W/K)



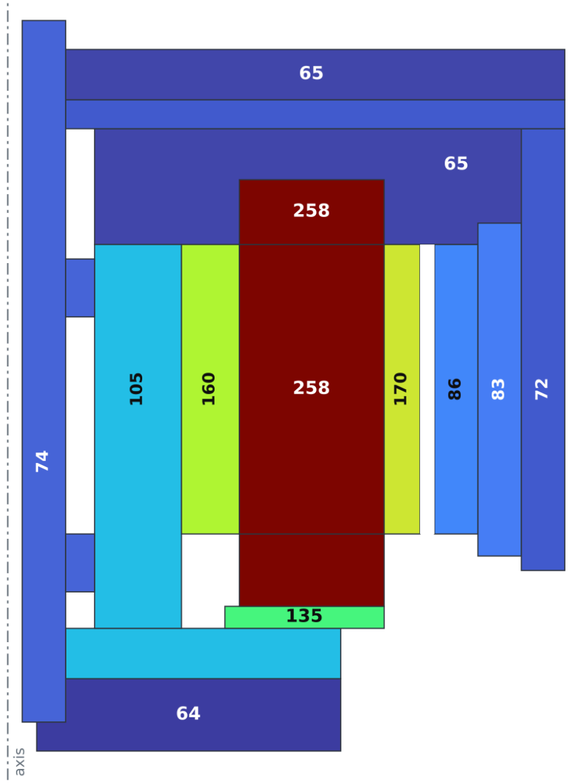
Same network as a graph, as drawn in the explainer. 27 conductances; convection terms scale with speed and prop downwash.

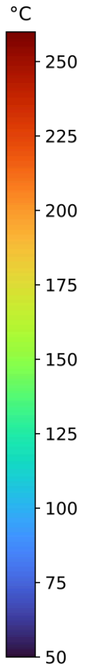
Where the heat sits after a 180 s race

Stock: end of 180 s race



Race build: end of 180 s race





258 °C

race-build end turns, at the 260 °C limit of its class 220 enamel

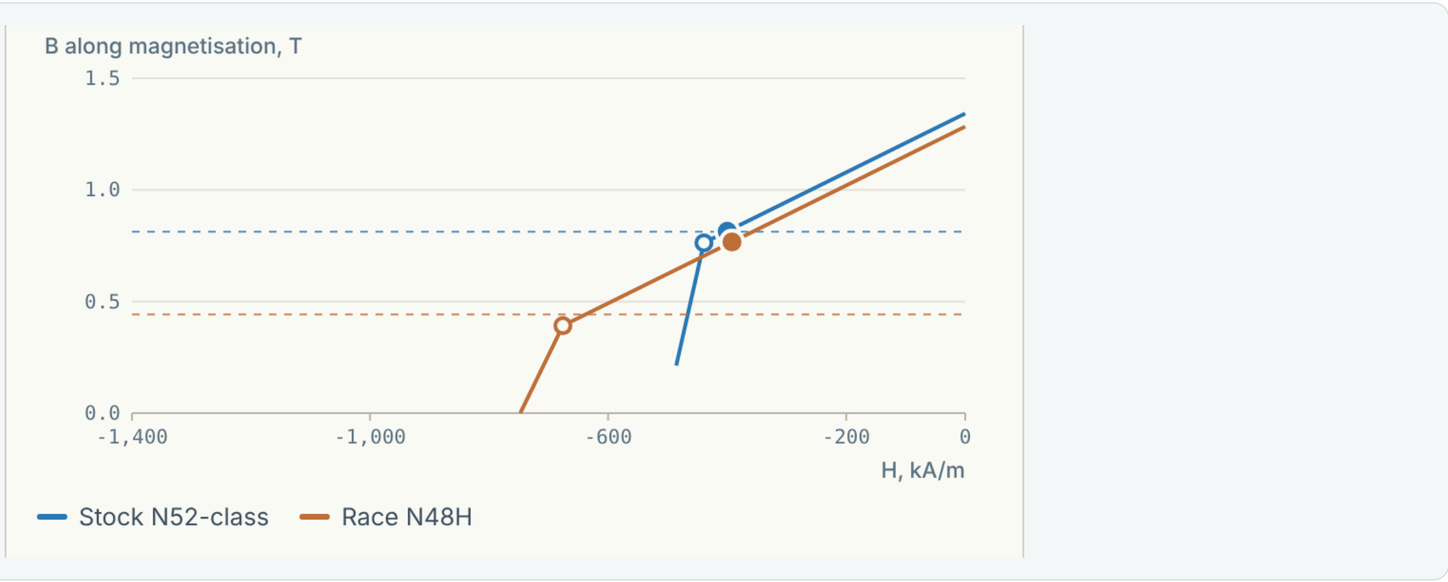
83 %

of the race heat is still stored in the motor

3.99 kW

at 98 A: matches the 4 kW, 180 s rating with no tuning

Four limits compete; the first one touched sets the power



Magnets row in detail: stock N52-class (blue) sits on the safety line; race N48H (orange) keeps 325 mT

Winding enamel	Class + 40 K for one race: 220 °C (class 180), 260 °C (class 220)
Magnets ◂ graph	Stock 2 mT above the 50 mT safety line at 79 °C; N48H keeps 325 mT at 86 °C
Printed parts	Stay 15 K under HDT: can at 72 °C against 135 °C allowed
Stress	Can 0.5 MPa; spider arm 172 MPa, so the spider

Three builds beat stock

Build	Key materials	Mass	180 s power	kW/kg	Limited by
Stock	NO20 0.20 mm, solid ring, N52-class, class 180, aluminium	1.78 kg	4.69 kW	2.63	magnet demag
Value	NO20, laminated ring, N48H, class 220, FDM PA6-CF can and cap	1.65 kg	4.98 kW	**3.03**	end plate
Race	NO10 0.10 mm, laminated ring, N48H, class 220, MJF PA12-CF, PEEK end plate	1.62 kg	5.01 kW	3.08	winding 260 °C
Budget retrofit	Stock stator, winding and magnets kept; laminated ring, FDM PA6-CF can and cap	1.65 kg	4.79 kW	2.90	magnet demag

Recommendation: build the value version first. It gets within 1.6 % of the race build for about €84 less, using the stock laminations and an FDM printer.

Budget retrofit: about €32 of new parts on an existing motor, no rewinding or new magnets. Good up to about 5-minute races; at 10 minutes the printed can traps rotor heat and it falls 14 % below stock.

360 material sets screened. Study v1: pre-teardown geometry, matched prop, 14S, 35 °C; masses ±10 %.

Short races are about mass, long races about heat paths

Up to about a minute

Heat has no time to leave the copper, so the copper's own heat capacity sets the current. Every build has the same copper, so power barely moves; only a lighter motor raises kW/kg.

A few minutes and up

Heat paths, magnet grade and the laminated ring decide it.

30 s, race vs stock

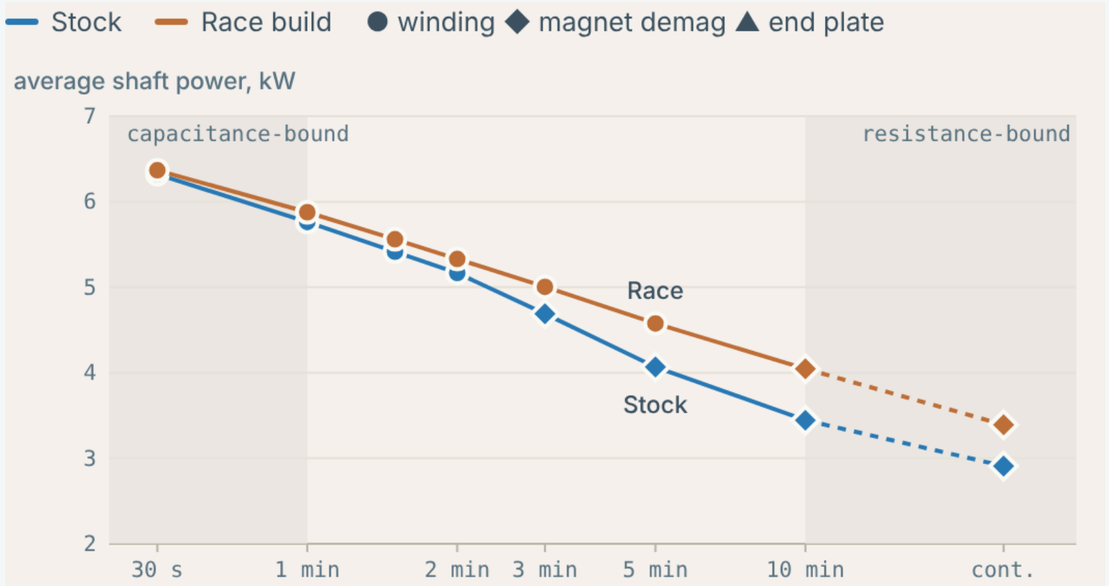
+10.5 % kW/kg

power only +0.7 %: the gain is the 9 % lighter motor

600 s, race vs stock

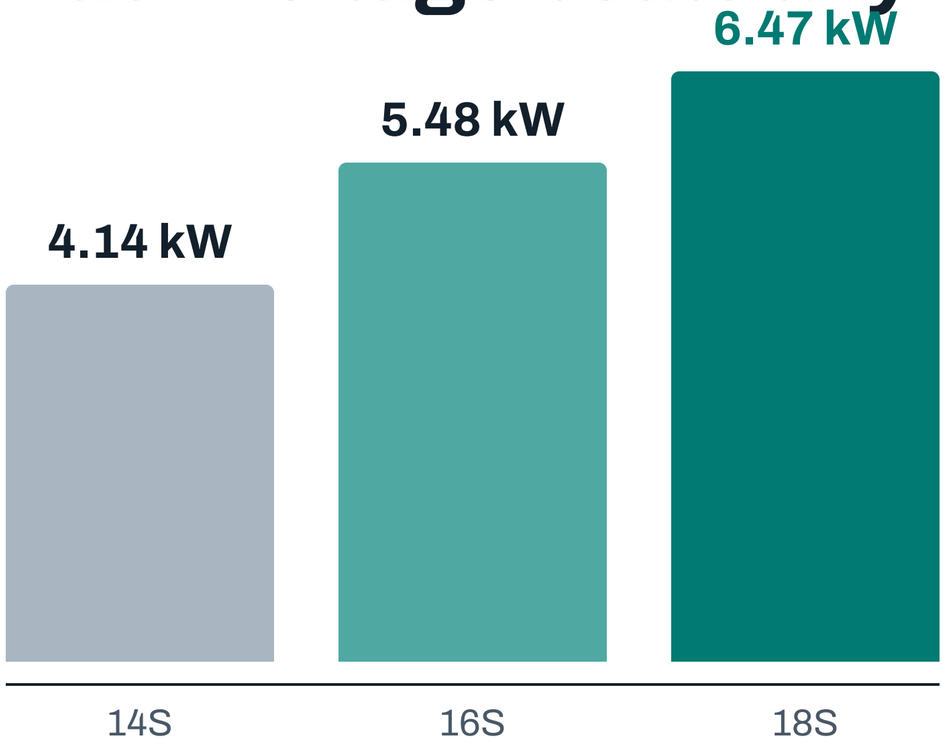
+29 % kW/kg

power itself +17.5 %, on top of the lighter motor



Average shaft power a race of each length can sustain (matched prop, 35 °C, from ambient). Marker shape = the limit that sets it.

Pack voltage beats any material change



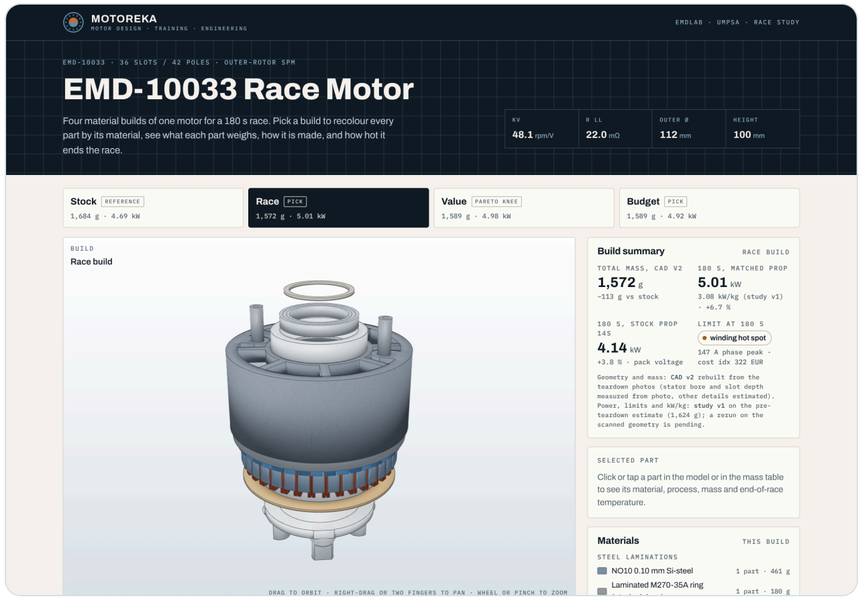
+56 %

Materials alone: +3.8 %. On the stock prop at 14S the pack voltage caps the speed, and a fixed prop takes power roughly as speed cubed. The race build's winding ends at only 125 °C against its 260 °C limit: the headroom goes unused.

14S to 18S: +56 %. More voltage lifts the speed cap; the prop spins faster until the winding reaches its limit.

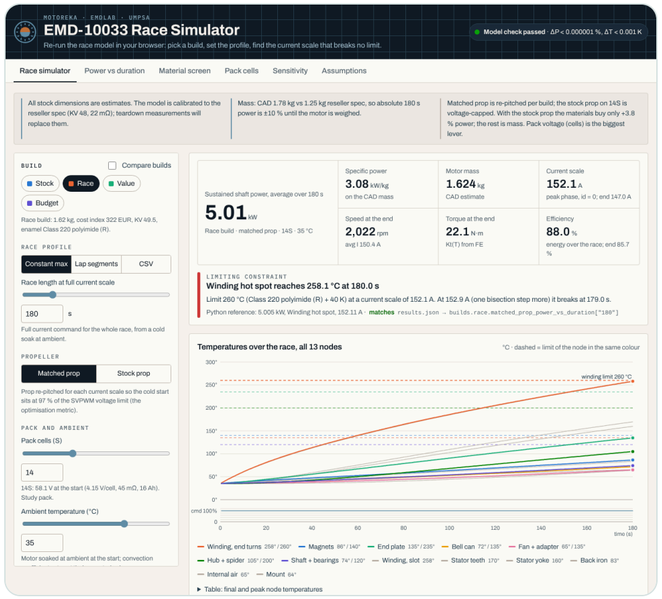
Race build, stock prop, 180 s. Needs a 70–80 V ESC and a rules check.

Every result is explorable in the browser



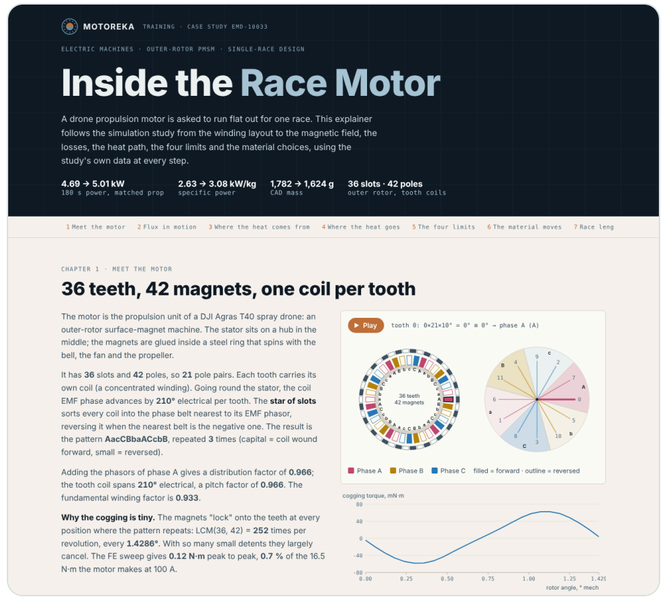
3D viewer

Orbit, explode and section the CAD; switch builds to recolour parts by material.



Race simulator

Set race length, pack and prop; see power and the limit that binds.



Animated explainer

Field, heat and material choices explained step by step for students.

What EMDLab can bring to UMPSA Aerospace

Motor selection and matching

Pick motor, prop and pack for a UAV mission, with thermal limits checked.

Reverse engineering

Teardown, 3D scan, CAD and refabrication data for propulsion motors.

Optimisation studies

Materials, windings and cooling against any mission or race profile.

Training and competitions

Drone propulsion modules and support for student race teams.

From model to a motor on the thrust stand

01

Calibrate on the bench

Weigh the parts; measure back-EMF, resistance and locked-rotor heating.

02

Rerun on scanned geometry

Fit CAD to the 3D scans and repeat the FE, thermal and material study.

03

Build and race-test

Print and assemble the value build, then run it on the thrust stand.

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